Critical Tree-Ising Information, Optimized Depth Filters, and Quantitative Innovation Tails

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Abstract

We study the observation and recovery mechanism in the September 26, 2026 manuscript on the sharp factor-of-IID threshold for the free Ising model. At the critical parameter, the exact covariance of normalized depth sums is a rank-one term plus a discrete Brownian covariance. Solving the resulting finite-depth optimization gives an explicit reflected geometric filter, its exact information value, and a relative truncation error. The infinite-depth bound improves the source's flat-cutoff information constant without changing its critical power. We transfer this improvement to intact-branch attenuation, derive an inverse-radius integrated response tail, and give a separate finite-boundary covariance-tail rate. Finite-volume causal residual estimates are stated with their volume-dependent constants. Neither these estimates nor their exact finite checks replace the source's critical spatial cancellation, compatible conditional-copy argument, or all-input equivariant spin selection. Contribution assessment: C3/5, the exact optimal depth filter and distinct quantitative spatial-tail budgets are notable local refinements, not an independent proof of the global factor threshold.

1 Model, status, and proof boundary

Let T_d be the infinite d-regular tree, $d \geq 3$, let $b = d - 1$, and let $\theta = \tanh \beta$ with $\beta \geq 0$. Its free zero-field Ising law is the broadcast law: a fair root sign is propagated by independent edge multipliers of mean θ. The auxiliary root does not enter the law. In particular

$$\mathbb{E}\sigma_u = 0, \qquad \mathbb{E}\sigma_u\sigma_v = \theta^{\mathrm{dist}(u,v)}.$$

An IID factor is a measurable spin map of independent uniform vertex labels, commuting almost surely with each fixed automorphism. The source [1] claims existence exactly for $\theta \leq b^{-1/2}$, including equality, and an everywhere equivariant Borel spin version. It does not claim finitary coding, a coding-radius tail, or uniform computation in d, β.

The distinction between a source claim and a result proved here is essential. We read the complete cached scientific manuscript and its ancillary text, but a literal reading is not independent certification of all stochastic, spectral, and measurable-selection arguments or of their external references. The local results below follow from explicit broadcast moments and, where specified, named source estimates. We do not use the claimed factor theorem to prove any of the moment optimizations.

On the planted probability space, sample σ with this law and independent standard Brownian motions B_v, independent also of σ, and set

$$X_v(t) = t\sigma_v + B_v(t), \qquad t \geq 0. \tag{1.1}$$

For a directed edge $v \to w$, its component $T_{v\to w}$ is the side containing v after deleting that edge. Its posterior uses the *unconditional free prior* on that component and only its own observation histories. Write

$$h_{v\to w}(t) = \operatorname{atanh} \mathbb{E}[\sigma_v \mid X_u(s) : u \in T_{v\to w},\, s \leq t], \qquad M(t) = \mathbb{E} \tanh^2 h_{v\to w}(t).$$

In the critical sections below $\theta = b^{-1/2}$ and $t > 0$. All statements about infinite random fields concern the planted law, not every continuous path array.

For a finite connected subtree D the posterior at terminal array x is

$$\pi_x^D(\eta) = Z_D(x)^{-1} \exp\left(\beta \sum_{\{i,j\}\in E(D)} \eta_i\eta_j + \sum_{i\in D} x_i\eta_i\right). \tag{1.2}$$

This follows by multiplying the broadcast prior by $\exp(\sum_i \eta_i X_i(t) - t|D|/2)$. Terminal sufficiency for the spins does not assert equality of terminal and path sigma-fields. Differentiation gives $\partial_{x_u} M_v^D = \operatorname{Cov}_{\pi_x^D}(\eta_v, \eta_u)$. The source's branch recursion and directed response are

$$\psi(z) = \operatorname{atanh}(\theta \tanh z), \qquad a(z) = \frac{1 - \tanh^2 z}{1 - \theta^2 \tanh^2 z}, \tag{1.3}$$

$$h_{v\to w} = X_v + \sum_{u\sim v,\ u\neq w} \psi(h_{u\to v}), \qquad c_{vu} = \theta^r \prod_{i=1}^{r} a(h_{v_i\to v_{i-1}}), \tag{1.4}$$

$$C_{vu} = (1 - \tanh^2 h_v) c_{vu}, \qquad A_v = \sum_u c_{vu}^2. \tag{1.5}$$

Here $v = v_0, \ldots, v_r = u$ is the unique path, and $c_{vv} = 1$. The covariance C is symmetric; the variance-normalized response c generally is not. Both $0 < a \leq 1$ and $|\psi'| \leq \theta$ are exact.

2 Exact depth covariance and the optimal linear statistic

Root the component at o. At its level k define

$$S_k = \theta^k \sum_{|u|=k} \sigma_u, \qquad Z_k = \theta^k \sum_{|u|=k} X_u(t), \qquad k \geq 0. \tag{2.1}$$

These are physical observations, not formal independent level variables. Let

$$c = \frac{b-1}{b}, \quad G_{k\ell} = 1 + c\min(k,\ell), \quad K_{k\ell} = \min(k,\ell) \quad (0 \leq k, \ell \leq n).$$

Lemma 2.1 (The exact Gram matrix)**.** *At criticality,*

$$\mathbb{E}[\sigma_o S_k] = 1, \qquad \mathbb{E}[S_k S_\ell] = G_{k\ell}, \qquad \operatorname{Cov}(Z_k, Z_\ell) = t\delta_{k\ell} + t^2 G_{k\ell}, \qquad \mathbb{E}[\sigma_o Z_k] = t.$$

Moreover K is positive semidefinite and $Q_n := I + ctK$ is positive definite with $Q_n \geq I$.

Proof. For $k \le \ell$, partition ordered pairs at levels k, ℓ by the depth j of their last common ancestor. For $j < k$ there are $(b-1)b^{k+\ell-j-1}$ pairs and each weighted covariance is $b^{-(k+\ell-j)}$, so their contribution is c. For $j = k$ the ancestor pairs contribute one. Summing gives $1+ck$. The covariance with the root follows by summing $b^k\theta^{2k} = 1$. Different levels use disjoint Brownian coordinates; each normalized Brownian level sum has variance t. This proves the identities. Finally

$$w^T K w = \sum_{j=1}^{n} \left(\sum_{k=j}^{n} w_k \right)^2 \ge 0.$$

□

For any nonzero real $w = (w_0, \dots, w_n)$, conditional expectation and Cauchy–Schwarz imply

$$M(t) \ge \frac{\left(\mathbb{E}[\sigma_o \sum_k w_k Z_k]\right)^2}{\mathbb{E}(\sum_k w_k Z_k)^2} = \frac{t(\mathbf{1}^T w)^2}{w^T(Q_n + t\mathbf{1}\mathbf{1}^T)w}. \tag{2.2}$$

Zero-sum weights give zero. The source uses $w_k = 1$ and an upper bound on its denominator. We retain the exact Gram matrix and optimize it.

Theorem 2.2 (Closed finite-depth optimum)**.** *Let $r \in (0,1)$ be the smaller positive solution of $r + r^{-1} = 2 + ct$, equivalently*

$$ct = \frac{(1-r)^2}{r}, \qquad 1 - r = \frac{\sqrt{c^2t^2 + 4ct} - ct}{2}. \tag{2.3}$$

For every integer $n \ge 0$ define

$$x_k = \frac{r^k + r^{2n+1-k}}{1 + r^{2n+1}} \quad (0 \le k \le n), \tag{2.4}$$

$$H_n = \frac{1 - r^{2n+2}}{(1-r)(1+r^{2n+1})}, \qquad I_n(t) = \frac{tH_n}{1 + tH_n}. \tag{2.5}$$

Then $Q_n x = \mathbf{1}$, $\mathbf{1}^T x = H_n$, and the supremum in (2.2) is $I_n(t)$, attained precisely by nonzero scalar multiples of x. Consequently $M(t) \ge I_n(t)$. The value is also optimal among all linear combinations of the individual terminal observations in the depth-n component ball.

Proof. The row $k = 0$ of $Q_n x = \mathbf{1}$ requires $x_0 = 1$. For $n \ge 1$, successive row differences require

$$x_k - x_{k-1} = -ct \sum_{\ell=k}^{n} x_\ell, \qquad 1 \le k \le n. \tag{2.6}$$

Their second differences are $x_{k+1} - (2+ct)x_k + x_{k-1} = 0$. The last row difference is $x_{n-1} = (1+ct)x_n$, or the reflecting condition $x_{n+1} = x_n$ in the recurrence. Formula (2.4) satisfies this recurrence, this boundary condition, and $x_0 = 1$. Backward summation proves (2.6) and hence all original rows. For $n = 0$, $Q_0 = x_0 = H_0 = 1$ directly. Geometric summation gives H_n.

Since $Q_n \ge I$, its inner product gives $(\mathbf{1}^T w)^2 = (x^T Q_n w)^2 \le H_n w^T Q_n w$. Insert this into (2.2); equality occurs precisely for w proportional to x. This proves the value without assuming an inverse on an infinite coefficient space.

For an arbitrary linear statistic in the finite component ball, average it over the finite group of rooted tree automorphisms. Its covariance with σ_o is unchanged, and its second moment cannot increase, by convexity and invariance. The averaged coefficients are constant on levels and therefore have the form (2.1) with level weights. A statistic with nonzero root covariance cannot average to zero. The strictly positive Brownian covariance also rules out a nonzero zero-variance statistic. Thus this averaging proves the last assertion. □

Corollary 2.3 (Infinite-depth value and exact finite error)**.** *The supremum over all finitely supported terminal linear statistics on the component equals*

$$I_\infty(t) = \frac{t}{t+1-r}, \qquad M(t) \geq I_\infty(t). \tag{2.7}$$

More precisely,

$$H_\infty = \frac{1}{1-r}, \qquad H_\infty - H_n = \frac{(1+r)r^{2n+1}}{(1-r)(1+r^{2n+1})}, \tag{2.8}$$

$$I_\infty - I_n = \frac{t(H_\infty - H_n)}{(1+tH_\infty)(1+tH_n)}, \qquad \frac{I_\infty - I_n}{I_\infty} \leq 2r^{2n+1}. \tag{2.9}$$

In particular $I_\infty(t) \sim \sqrt{t/c}$ *as* $t \downarrow 0$*. For* $0 < t \leq 1$ *and for* $t \geq 1$*, respectively,*

$$I_\infty(t) \geq \kappa_c\sqrt{t}, \qquad I_\infty(t) \geq \kappa_c, \qquad \kappa_c = \frac{1}{1 - c/2 + \sqrt{c + c^2/4}}. \tag{2.10}$$

Proof. The formula for H_n increases to H_∞. Subtract it to obtain (2.8), and apply the increasing function $u \mapsto tu/(1+tu)$. Division by I_∞ gives a relative error at most $(1+r)r^{2n+1}/(1+r^{2n+1}) \leq 2r^{2n+1}$. Every finite support lies in some ball, which proves the supremum. No unbounded infinite matrix has been inverted.

Writing

$$\frac{I_\infty(t)}{\sqrt{t}} = \frac{1}{(1-c/2)\sqrt{t} + \sqrt{c + c^2 t/4}}$$

proves the first lower bound and the small-time asymptotic. Also $I_\infty(t) = [1 - c/2 + \sqrt{c^2/4 + c/t}]^{-1}$ is increasing in t. This proves the second bound and the value at one. □

The original lower bound is $\min(\sqrt{t}, 1)/5$. Here $\kappa_c > 1/2$ for $1/2 \leq c < 1$, so the improved uniform constant is more than twice as large; the asymptotic coefficient is $c^{-1/2}$. These are lower bounds on the nonlinear posterior information, not an assertion that $M(t) = I_\infty(t)$. The matrix method is classical linear prediction. Its concrete increment here is the exact tree covariance, solved reflecting boundary condition, and source-specific finite-depth error.

3 Worked depth certificate and implementation costs

Take $b = 2$, $\theta = 1/\sqrt{2}$, $c = 1/2$, and $t = 1$. Then $r = 1/2$. All matrices and optimal depth weights are rational, despite the irrational edge parameter:

n	x	H_n	I_n
0	(1)	1	1/2
1	(1, 2/3)	5/3	5/8
2	(1, 6/11, 4/11)	21/11	21/32
∞	$(2^{-k})_{k\geq 0}$	2	2/3

At $n = 2$ the exact error is 1/96, and the relative error is 1/64. Formula (2.9) bounds it by 1/16. The source's crude flat-cutoff lower bound at this time is 1/5; using its own exact flat statistic with $n = 2$ instead gives 18/29. Thus the gain is not obtained merely by comparing against an intentionally loose rounded constant.

For a relative tolerance $\varepsilon \in (0,1)$ it suffices that $2n+1 \geq \log(2/\varepsilon)/(-\log r)$. Since $-\log r \sim \sqrt{ct}$, depth of order $t^{-1/2}\log(1/\varepsilon)$ suffices for this *linear* information value. Generating $n+1$ weights costs $O(n)$ arithmetic operations after r is known. Collecting the actual observations through depth n uses $(b^{n+1}-1)/(b-1)$ vertices. We assert neither a measured runtime improvement nor finite-radius recovery of the spins. At small t, evaluating $1-r$ by subtraction can lose digits; the equivalent positive formula

$$1 - r = \frac{2ct}{\sqrt{c^2t^2 + 4ct} + ct}$$

avoids that subtraction. Rational choices of r and $t = (1-r)^2/(cr)$ give exact reproducible test fixtures.

3.1 A physical error certificate for computed weights

For weights normalized by $\mathbf{1}^T w = 1$, the information certificate is $J(w) = t/(t + w^T Q_n w)$. If $w_* = x/H_n$ and $e = w - w_*$, then

$$w^T Q_n w = H_n^{-1} + e^T Q_n e, \qquad I_n - J(w) = \frac{t\, e^T Q_n e}{(t + H_n^{-1})(t + H_n^{-1} + e^T Q_n e)}. \tag{3.1}$$

Indeed $Q_n w_* = \mathbf{1}/H_n$ and $\mathbf{1}^T e = 0$, so the cross term vanishes. The penalty has a directly interpretable decomposition

$$e^T Q_n e = \sum_{k=0}^{n} e_k^2 + ct \sum_{j=1}^{n} \left(\sum_{k=j}^{n} e_k \right)^2 .$$

It measures both independent observation noise and accumulated spin-level correlations; replacing it by an unweighted coefficient norm would discard the latter. If z is a computed unnormalized solution and $\rho = Q_n z - \mathbf{1}$, then $Q_n \geq I$ gives $\|z - x\|_2 \leq \|\rho\|_2$ and $|\mathbf{1}^T z - H_n| \leq \sqrt{n+1}\|\rho\|_2$. This is a classical residual bound in the actual observation metric, consistent with the original finite-inverse proof in [7]. It is not credited as a new inversion principle.

4 Improved attenuation and an infinite-response tail

This section uses the source's component posterior and conditional independence identities, but supplies explicit constants and the quantitative integration. Set

$$c_\beta = \min\left\{ \left(\frac{\tanh(2\beta+1)}{2\beta+1} \right)^2, \frac{\tanh^2(1)}{\beta^2} \right\}, \quad \gamma = \min\{1/2, c_\beta(1-\theta^2)^2\theta^2\}, \quad \eta = \gamma\kappa_c. \tag{4.1}$$

At criticality $\beta > 0$, so these constants are positive and finite.

Proposition 4.1 (An explicit improved path loss)**.** *For a deterministic simple path $v_0, \ldots, v_r$ and $t > 0$,*

$$\mathbb{E}\left[\prod_{i=1}^{r} a(h_{v_i \to v_{i-1}}(t))^2 \,\middle|\, \sigma_{v_0}, \ldots, \sigma_{v_r} \right] \leq (1 - \gamma I_\infty(t))^r \leq e^{-\eta \min(\sqrt{t},1) r}. \tag{4.2}$$

The reverse-oriented product obeys the same estimate.

Proof. Let Y be a component field and $G = \psi(Y)$ the outgoing message. Its marginal law is symmetric and $|G| \le \beta$. Given the attaching spin $\sigma_w = s$, its density relative to its marginal is $1 + s\theta \tanh Y$. Even functions retain their expectations. Consequently

$$\operatorname{Var}(G \mid \sigma_w = s) \ge (1-\theta^2)\mathbb{E}G^2 \ge (1-\theta^2)\theta^2 M(t).$$

The first inequality follows by Cauchy–Schwarz on $\mathbb{E}[G \tanh Y]$, and the second from $|\operatorname{atanh} u| \ge |u|$.

Uniformly over every real shift z and every law of $G \in [-\beta, \beta]$, $\mathbb{E}\tanh^2(z+G) \ge c_\beta \operatorname{Var} G$. For $|z| \le \beta+1$, use the minimum of $\tanh^2 u/u^2$ on $|u| \le 2\beta+1$ and $\mathbb{E}(z+G)^2 \ge \operatorname{Var} G$. The ratio $\tanh u/u$ decreases for $u > 0$: the derivative numerator $u \operatorname{sech}^2 u - \tanh u$ starts at zero and has negative derivative. For $|z| > \beta + 1$, use $|z+G| > 1$ and $\operatorname{Var} G \le \beta^2$. Finally $a(h)^2 \le 1 - (1-\theta^2)\tanh^2 h$ yields

$$\mathbb{E}[a(z+G)^2 \mid \sigma_w = s] \le 1 - \gamma M(t) \le 1 - \gamma I_\infty(t).$$

Select at each target of the path an intact infinite off-path component. At least one exists, including when $d = 3$. Conditional on the path spins and the outside observations, these components are independent with the attaching-spin laws just used. Integrate them in order from the initial target toward the final one. A later target field excludes earlier selected components, so each integration removes one factor with the displayed loss. This is the directed independence mechanism of the source, not independence of all fields along the path. Reversal gives the opposite orientation. Now apply (2.10) and $1 - u \le e^{-u}$. □

Theorem 4.2 (Integrated spatial tail for the actual infinite row)**.** *Assume the infinite coefficients* (1.4) *under the planted law, with their predictable representatives when integrated. For $T > 0$, integer $L \ge 1$, and $T_1 = \min(T, 1)$, put*

$$\mathcal{R}_L(T) = \mathbb{E}\int_0^T \sum_{\operatorname{dist}(v,u)>L} c_{vu}(s)^2 \, \mathrm{d}s.$$

Then

$$\mathcal{R}_L(T) \le \frac{2d}{b\eta^2 L}(1 - e^{-\eta L\sqrt{T_1}}) + \frac{d}{b}(T-1)_+ \frac{e^{-\eta L}}{e^\eta - 1}. \tag{4.3}$$

In particular $\mathcal{R}_L(T) = O_{d,T}(L^{-1})$. The same bound holds for the covariance coefficients C_{vu}. If W is Brownian in the observation filtration, then

$$\left\| \sum_{\operatorname{dist}(v,u)>L} \int_0^\cdot c_{vu}(s)\, \mathrm{d}W_u(s) \right\|_{L^2(\Omega; C[0,T])} \le 2\sqrt{\mathcal{R}_L(T)} = O_{d,T}(L^{-1/2}). \tag{4.4}$$

Proof. The size of shell r is db^{r-1} and its squared critical edge factor is b^{-r}. Tonelli and (4.2) give

$$\mathbb{E}\sum_{\operatorname{dist}(v,u)>L} c_{vu}(s)^2 \le \frac{d}{b}\,\frac{e^{-\eta\omega(s)L}}{e^{\eta\omega(s)} - 1}, \qquad \omega(s) = \min(\sqrt{s}, 1).$$

For $s \le 1$, the denominator is at least $\eta\sqrt{s}$. Substitute $y = \sqrt{s}$ and integrate the exponential exactly. For $s > 1$ the bound is constant in s, giving the second term. Multiplication by $(1 - m_v^2)^2 \le 1$ proves the covariance statement. For finite annuli, Itô isometry and Doob's inequality give the square bound $4\mathcal{R}_L(T)$. The integrated tail tends to zero, so these martingales converge in maximal L^2 and the inequality passes to their limit. □

This is a truncation of the *stochastic sum*, not a localization of its coefficients. Even c_{vu} for nearby u uses infinite component observations. At time zero $a(0) = 1$ and the critical untruncated row is infinite. Formula (4.3) integrates across that single exceptional time; it does not supply a finite row bound at zero. The information improvement strengthens the explicit constant η, but the L^{-1} order already follows by quantitatively integrating the source's intact-branch attenuation.

5 A separate uniform finite-boundary covariance budget

The preceding proof cannot be transplanted to a boundary leaf: there is no intact infinite side component there. The source uses the target's own Gaussian observation instead. This section makes its two-limit estimate into a single explicit budget.

Let $D_R = B_R(v)$ with free boundary, and extend its covariance $C_{vu}^{D_R}$ by zero outside D_R. Define

$$\mathcal{E}_L(T) = \sup_R \mathbb{E} \int_0^T \sum_{\operatorname{dist}(v,u)>L} (C_{vu}^{D_R}(s))^2 \, \mathrm{d}s.$$

The required source inputs are its finite posterior energy identity and its critical information upper estimate. They concern the true finite posteriors under the same planted law, not a fitted covariance matrix.

Proposition 5.1 (Uniform covariance tail with an explicit rate)**.** *Set* $a_0 = \sqrt{2\pi}/4$, *and*

$$c_G = \frac{1-\theta^2}{2} \tanh^2 a_0, \qquad K_d = 2 + \frac{d}{\sqrt{c}}, \qquad A = \frac{2}{c_G}.$$

For $N = L + 1 \geq 2$, *if* $\delta = A \log N/N \leq \min(1, T)$, *then*

$$\mathcal{E}_L(T) \leq K_d \sqrt{\frac{A \log N}{N}} + \frac{dT}{2bN \log N}. \tag{5.1}$$

The same bound holds for the true infinite covariance tail. Thus it is $O_{d,T}(\sqrt{\log L/L})$, *uniformly in the finite-ball radius. This assertion is weaker than* (4.3) *for the infinite row, but has a stronger finite-boundary quantifier.*

Proof. We first track the early-time constant. The source's binary-signal fusion identity for independent signals conditional on a fair spin subtracts

$$\mathbb{E}_{\mathrm{prod}} \frac{U^2V^2(2 - U^2 - V^2)}{1 - U^2V^2}.$$

For two child signals $|U|, |V| \leq \theta$ the loss is at least $2(1-\theta^2)\theta^4 M(t)^2$. Combining those two first, then using subadditivity for the remaining signals, gives $2cb^{-2}M(t)^2 \leq q_0(t)$ at criticality. Here $q_0(t) \leq t + t^2 \leq 2t$ for $t \leq 1$. Hence $M(t) \leq b\sqrt{t/c}$ and the whole-site information is at most $q_0(t) + (d/b)M(t) \leq K_d\sqrt{t}$. Conditional Jensen and finite posterior energy therefore give

$$\mathbb{E} \int_0^\delta \sum_u (C_{vu}^{D_R}(s))^2 \, \mathrm{d}s = \mathbb{E}(m_v^{D_R}(\delta))^2 \leq K_d\sqrt{\delta}.$$

Choose $\varepsilon_\delta = a_0\sqrt{\delta}$. Every Gaussian of variance at least δ, with any mean, assigns mass at most $1/2$ to $[-\varepsilon_\delta, \varepsilon_\delta]$. The inequality for a^2 used above and the source's ordered integration of the target Gaussian noises imply

$$\mathbb{E}(C_{vu}^{D_R}(s))^2 \leq \theta^{2\operatorname{dist}(v,u)}(1-\kappa_\delta)^{\operatorname{dist}(v,u)}, \qquad \kappa_\delta = \frac{1-\theta^2}{2} \tanh^2(a_0\sqrt{\delta}).$$

This integration works at leaves. Concavity of tanh on the positive half-line implies $\kappa_\delta \geq c_G\delta$ for $\delta \leq 1$. Summing the critical shells on $[\delta, T]$ and adding the early energy yields

$$\mathcal{E}_L(T) \leq K_d\sqrt{\delta} + \frac{dT}{b\, c_G\delta} e^{-c_G\delta N}.$$

The indicated δ makes the exponential equal N^{-2}, proving (5.1). For the infinite covariance row, first pass to the limit on a finite annulus using bounded posterior convergence and then enlarge the annulus by monotone convergence. No rate for the finite-coordinate convergence is assumed. □

For a finite set $F = B_L(v)$ and approximations $\widehat{C}_{vu}$, assume both coefficient families are square-integrable and predictable in a common filtration where the same W is Brownian. Itô isometry and the disjoint noise coordinates give the exact coefficient error decomposition

$$\mathbb{E}\left|\sum_u \int_0^T C_{vu}\,\mathrm{d}W_u - \sum_{u\in F}\int_0^T \widehat{C}_{vu}\,\mathrm{d}W_u\right|^2 = \mathbb{E}\int_0^T \sum_{u\notin F} C_{vu}^2\,\mathrm{d}s + \mathbb{E}\int_0^T \sum_{u\in F} |C_{vu} - \widehat{C}_{vu}|^2\,\mathrm{d}s. \tag{5.2}$$

The maximal squared error is at most four times this expression. Finite posterior innovations W^{D_R} are not the infinite W. Substituting them into the second integral changes the problem and requires an additional driving-process comparison. This is why the tail rates are not, by themselves, local IID simulation algorithms.

6 Causal inversion: what finite volume actually certifies

The freshly read original Ram Urysohn–Volterra proof [6], equations labelled `eq:volterra` and `eq:bielecki`, separates source-state feedback from instantaneous observation-state feedback. In its notation, a source-state Lipschitz constant is reduced by an exponential time weight, whereas an observation-state constant contributes BT. The example $f(t) = 1 + tf(t)$ prevents unrestricted claims based on causality alone. The source-only finite-dimensional posterior equation below fits the former case, but with a spatially growing constant.

Proposition 6.1 (Finite-volume all-input residual estimate)**.** *For a finite connected subtree D, put $L_D = \max_{v\in D}\sum_{u\in D}\theta^{\mathrm{dist}(v,u)}$. For every continuous path $w : [0,T] \to \mathbb{R}^D$ with $w(0) = 0$, the equation*

$$x(t) = w(t) + \int_0^t M^D(x(s))\,\mathrm{d}s \tag{6.1}$$

has a unique continuous solution. For $\lambda > L_D$, define $\|z\|_\lambda = \sup_{t\leq T} e^{-\lambda t}\|z(t)\|_\infty$. For any continuous candidate z and residual $R_z = z - w - \int_0^{\cdot} M^D(z(s))\,\mathrm{d}s$,

$$\|z - x\|_\lambda \leq \frac{\|R_z\|_\lambda}{1 - L_D/\lambda}. \tag{6.2}$$

Picard iteration has contraction factor at most L_D/λ.

Proof. The finite posterior derivative and the tree product imply $|\partial_{x_u} M_v^D| \leq \theta^{\mathrm{dist}(v,u)}$ for every real x. The mean-value theorem gives global ℓ^∞ Lipschitz constant L_D. Also $|M_v^D| \leq 1$, so the integral map is everywhere defined and preserves continuous paths starting at zero. Its difference is bounded in the weighted norm by

$$L_D \sup_{t\leq T}\int_0^t e^{-\lambda(t-s)}\,\mathrm{d}s\,\|x - y\|_\lambda \leq (L_D/\lambda)\|x - y\|_\lambda.$$

This norm is equivalent to the ordinary supremum norm on the finite horizon, so completeness and the contraction theorem apply. Subtract the exact equation from the candidate equation and absorb the contraction term to prove the residual bound. □

For a radius-R ball at the origin of its terminal field space, $\partial_{x_u} M_o^{D_R}(0) = \theta^{\mathrm{dist}(o,u)}$. The root row has norm

$$1 + d\theta \sum_{j=0}^{R-1} (b\theta)^j. \tag{6.3}$$

At criticality $b\theta = \sqrt{b} > 1$, so even this one row diverges exponentially in R. A uniform terminal-array supremum-norm Lipschitz constant cannot follow from these finite posteriors. Choosing a time weight λ_R larger than the diverging constant does not produce a fixed infinite-volume norm: conversion back to ordinary path error costs $e^{\lambda_R T}$. Further, a countable array of independent Brownian coordinates is not uniformly bounded over all vertices at a positive time. One must declare a genuine infinite path space instead of silently using $C([0,T];\ell^\infty)$.

The source deliberately follows a different route. Its observation space is the countable product of $C_0([0,\infty),\mathbb{R})$ with locally uniform coordinate topology. It proves equality of two observations sampled conditionally on their common innovations, under a specific invariant law. This is not uniqueness for every input in an unrestricted infinite path equation. The finite causal proposition therefore remains a finite-volume error certificate only.

7 Critical cancellation and recovery obligations

Here we give the full dependency map, while retaining the distinction between checked local deductions and the source's global mechanism. The source's nine scientific modules have the following roles.

Module	Essential content and obligation
01 Introduction	The precise threshold claim, BSV strict obstruction, no finitary claim, and almost-sure versus everywhere equivariance.
02 Posterior fields	Free finite posterior, unconditional component prior, path covariance, progressive Borel versions, and directed normalization.
03 Critical information	Binary fusion, depth lower bound, infinite-side attenuation, and integrable response rows. Our exact depth optimization refines this lower-bound step.
04 Innovations	Brownian innovations in raw and usual filtrations, finite energy, uniform finite-boundary covariance tails, exact limiting energy, no residual orthogonal martingale, and continuous local versions.
05 Coupling	Conditional copies given the entire innovation paths, past compatibility, transported Itô integrals, discrepancy noise *and* drift, and strict-subcritical uniqueness.
06 Spectrum	One-copy vertex spectrum, its transfer to the closed space of marginal differences, nonbacktracking powers, and linear path variance.
07 Path averages	Both orientations, cutting the incoming message at bounded cost, conditional descendant concentration, and dependent weights.
08 Critical comparison	All-order small-field moments, discrepancy-dependent spatial cutoff, integrable Osgood comparison, and critical conditional-copy equality.

Module	Essential content and obligation
08b Recovery	Conditional point-mass recovery on countably many rational coordinates, spin limits, uniform-to-Wiener sampler, and everywhere equivariant spin modification.

7.1 Why square-row bounds are insufficient

Under the compatible joining, write $V(t) = \mathbb{E}(h_o - h'_o)^2$ and $S(t) = \mathbb{E}(\log a(h_{j\to i}) - \log a(h'_{j\to i}))^2$. The source uses the exact discrepancy identity

$$V' = \sum_u \mathbb{E}(c_{ou} - c'_{ou})^2 + 2\mathbb{E}[(h_o - h'_o)(m_o A_o - m'_o A'_o)].$$

At criticality the shell factor is $db^{r-1}\theta^{2r} = d/b$. Crude path Cauchy–Schwarz gives $r^2 S$, which is insufficient. The source proves $O(rS)$ on the closed linear space of differences $g(X) - g(X')$, not on every function of the pair. Even combined with attenuation, this yields the leading term $S(t)/t$. The elementary bound $S \le CV$ would leave a nonintegrable coefficient at zero. For instance the scalar function $V(t) = t$ has $V(0) = 0$ and $V' = V/t$ for $t > 0$, so that inequality cannot force zero.

The source instead proves

$$S(t) \le C_T V(t) \left(t^{1/4} + \frac{1 + \log(K_T/V(t))}{\log^2(1/t)} \right)$$

at sufficiently small positive times where $V > 0$. It also controls the square-root path sum in the drift. These give an Osgood inequality with time coefficient bounded by a multiple of $1 + t^{-3/4} + [t \log^2(1/t)]^{-1}$ near zero, which is integrable. Our new lower bound on M and tail integrations do not replace the all-moment tail, marginal-difference spectrum, dependent path-weight concentration, or this final nonlinear comparison.

7.2 Measurable recovery is not a time integrator

Let $\mathsf{K}(w, \mathrm{d}x)$ be the observation conditional law given the entire innovation array. The source constructs the joining $\mathsf{w}(\mathrm{d}w)\mathsf{K}(w, \mathrm{d}x)\mathsf{K}(w, \mathrm{d}x')$. Brownianity in the joined past is a proved compatibility condition, not a consequence of the two marginal Brownian laws. If the source comparison gives $X = X'$ there, then for every vertex and rational q,

$$\mathbb{E}[(X_v(q) - X'_v(q))^2 \mid W] = 2\,\mathrm{Var}(X_v(q) \mid W) = 0.$$

Countably many rational coordinates determine continuous paths, so the conditional law is a point mass almost surely. The Borel reconstruction uses dyadic interpolants and explicit defaults. Under that reconstruction $X_v(n)/n \to \sigma_v$ almost surely, since $\sum_n \mathbb{P}(|B_v(n)| > \varepsilon n) < \infty$. This proves a measurable recovery statement only after all previous gates hold.

Finally one cannot intersect null sets over all automorphisms, an uncountable group. The source averages each finite-ball conditional spin expectation over the finite rooted-ball automorphism group, uses the same rule at every vertex, and takes a limsup followed by a common sign threshold. Each finite rule and its final limsup are equivariant on every input. Convergence to the recovered spin holds almost surely by martingale convergence. The explicit threshold is important; a conditional expectation is not itself a spin. The relevant published erratum of Lyons [5] similarly distinguishes conditional expectations from their indicator-valued approximations.

8 Comparisons, literature, and original Ram inputs

Public primary records were refreshed on October 9, 2026, using public titles and URLs only. Nam–Sly–Zhang [2] construct the free Ising factor in a constant fraction of the reconstruction scale; the source specifies a sufficiently large degree condition for that theorem. Their published result is not cited as proving the full endpoint asserted by the September 2026 source. The BSV paper [3] provides the correlation obstruction above the reconstruction threshold. Kurtz [4] treats compatibility in weak/strong stochastic solution principles. Those public abstracts and version records were checked here; their complete proofs were not independently audited. The primary search also located the release's own formalization-scope page. We did not inspect or execute a Lean proof, and do not use that page as independent mathematical validation of the source.

The private original TR6 passages freshly read were lines 759–975: causal feedback, weighted contraction, regular and singular Volterra differentiation, causal tangents, exact finite moments, and the limits of truncated response expansions. The original TR1 stability module, lines 1–125, supplies the finite perturbation/residual proof and the requirement to use the physical observation covariance. Their role is methodological and precise: keep time inversion separate from space, retain the full Gram metric, and distinguish a finite response calculation from an infinite inverse. We do not claim these originals proved the tree factor theorem, the depth formulas above, or the critical path cancellation. Their remaining passages and all other private corpus files were not freshly read for this report.

Object	Source or classical input	Increment proved here
Component information	Flat depth statistic; critical $\sqrt{t}$ scale	Optimal finite weights, reflecting endpoint, exact value and relative depth loss
Infinite response row	Intact-branch attenuation; integrability	Explicit improved constant and L^{-1} integrated tail
Finite-boundary covariance	Own-Gaussian attenuation and early energy	Uniform single-cutoff $\sqrt{\log L/L}$ budget
Causal inversion	Classical Volterra contraction and finite posterior derivatives	Physical residual statement and explicit reason its spatial constant diverges
Global factor	Compatible joining, spatial cancellation, Osgood and Borel selection	No independent replacement or new global theorem

9 Costs, limitations, and next mathematical questions

The principal advantage is an exact, positive depth design rather than a generic dense matrix overlay. Its finite solution has a reflecting endpoint; using the infinite geometric weights at the last level without that correction would not solve the finite normal equations. The improved information lower bound propagates through the same conditional side-branch proof. The two spatial budgets specify which row and which boundary quantifier they actually control, and (5.2) retains the unresolved nearby-coefficient error.

The costs are equally concrete. The number of vertices grows exponentially in depth. The nonlinear posterior may contain more information than every linear statistic considered here. Constants in the attenuation estimate need not be efficient. The finite-boundary rate requires the stated positive cutoff to lie below both 1 and T; it is an asymptotic bound with an explicit entry condition, not a small-L assertion. The infinite-row Doob estimate does not control the nonlinear field drift error, the error of a numerical stochastic integral, or a change of driving filtration.

A useful next problem is a quantitative finite-coordinate posterior convergence bound compatible with the uniform finite energy tails. Only then could one attempt a full observable approximation with a declared error in the coefficients, the noise, and the drift. Another is to extend the depth design to nonconstant branching or observation variances; the symmetric normal equations would change and no regular-tree formula should be transplanted unchanged. Finally, sharper binary-fusion upper information estimates could compare the nonlinear posterior with the optimal linear value. None of these prospective developments is a conclusion of this report.

10 Evidence and conclusion

The read ledger records every displayed source interval with pre-read and post-read SHA-256 hashes. All nine scientific modules, the main file, figure, bibliography, and README were read literally. The private read was limited to the original intervals specified above. A source hash or heading map does not count as a proof read. The separate source-identity evidence compares two captured trees; it makes no live-branch or mathematical-validity assertion.

The accompanying owned exact checker verifies finite tree last-common-ancestor counts, rational finite normal equations, the exact and relative truncation formulas, the normalized weight-regret identity, and elementary covariance-tail budget inequalities in declared finite fixtures. The executed check passed 957 parameter cases and 9,229 exact rational assertions; case and assertion counts are also separated by family in its receipt. No external or parent checker is counted as author execution. These tests supplement the proofs; they do not certify the infinitely many input paths, all spatial levels, or the global factor statement.

The local mathematical conclusion is therefore specific. The source's depth statistic admits a solved optimal replacement, with a larger critical information constant and an exact finite-depth certificate. Its attenuation mechanisms also give two useful but inequivalent spatial-tail rates. These refinements justify C3 as a notable local contribution, with classical prediction, concentration, and stochastic integration tools credited. Applicability L3 remains a separate catalogue assessment.

References

[1] OpenAI, *The sharp factor-of-IID threshold for the free Ising model on regular trees*, September 26, 2026, pinned manuscript, public source directory. Full cached scientific text read; global result treated as the source's claim, not independently certified here.

[2] D. Nam, A. Sly, and L. Zhang, *Ising model on trees and factors of IID*, Commun. Math. Phys. 389 (2022), 1009–1046, arXiv:2012.09484v2 (January 22, 2022 version).

[3] A. Backhausz, B. Szegedy, and B. Virág, *Ramanujan graphings and correlation decay in local algorithms*, Random Structures Algorithms 47 (2015), 424–435, arXiv:1305.6784.

[4] T. G. Kurtz, *Weak and strong solutions of general stochastic models*, Electron. Commun. Probab. 19 (2014), no. 58, arXiv:1305.6747v2.

[5] R. Lyons, *Erratum to Factors of IID on Trees*, April 10, 2020, author's one-page erratum. Full erratum read; not a replacement audit of the original paper.

[6] M. Subbarao, original confidential Ram corpus, *TR6-lin-non-lin.tex*, RamNonlin-IE-PDE-astra, lines 759–975, especially equations `eq:volterra`, `eq:bielecki`, and `eq:causaltangent`. Local original proof passages only; no public upload.

[7] M. Subbarao, original confidential Ram corpus, *TR1-lin-nonlin/stability.tex*, lines 1–125, especially `thm:pert` and its residual variant. Finite scaled metric, not infinite-volume coercivity.